# RYAN WILSON

Security Engineer

Des Moines, IA (open to remote)

ryan@ryanwilson.io | ryanwilson.io | github.com/rywils | linkedin.com/in/rywils

## SUMMARY

Security-focused engineer with 8+ years of experience across networking, systems administration, and infrastructure operations. Specialized in enterprise networking, Linux systems, and cloud environments (AWS, Azure), with hands-on experience in Fortinet NGFW, Cisco Firepower, CrowdStrike Falcon, and Snort. Currently pursuing a B.S. in Cybersecurity and Information Assurance while developing Bitprobe, a vulnerability scanning platform for web, network, and TLS analysis. Focused on security engineering, infrastructure security, and AI-driven security systems.

## TECHNICAL SKILLS

**Security:**
Vulnerability assessment, threat detection, incident response, network segmentation, Fortinet NGFW, Cisco Firepower, Snort, CrowdStrike Falcon, Splunk, Wireshark, Nmap, Burp Suite

**Networking:**
TCP/IP, VLANs, VPNs, routing & switching, DNS, DHCP, Cisco, Juniper, Aruba, pfSense, Riverbed

**Systems:**
Linux administration, Windows Server, Active Directory, system hardening, VMware, Proxmox, Hyper-V, Ubuntu, RHEL, Debian, Kali, AlmaLinux, SELinux

**Cloud / Infrastructure:**
AWS, Azure, GCP, Terraform, Ansible, Docker, Kubernetes, VPC networking, WAF, CloudFront/CDN, CloudWatch

**Development:**
JavaScript/TypeScript, Python, Go, C, C#, PHP, Node.js, React, REST APIs, GraphQL, Supabase, PostgreSQL, MySQL, MariaDB

## PROFESSIONAL EXPERIENCE

### Freelance Developer

*ryanwilson.io | Nov 2024 - Present*

- Develop web applications using React and Next.js, along with WordPress sites for small business clients
- Project specific, full-stack applications using Next.js, Vue/Nuxt, SvelteKit, and Laravel
- Build backend APIs and services using Node.js, Express, GraphQL, tRPC, and Supabase
- Deploy applications across AWS, Vercel, Netlify, Linode, and DigitalOcean
- Implement CI/CD pipelines, CDN distribution, and Redis caching with Upstash
- Design application data layers using PostgreSQL, SQLite, MongoDB, and MariaDB
- Manage full project lifecycle including client onboarding, architecture, deployment, and support

### Systems Engineer

*DH Pace | Jul 2023 - Aug 2024*

- Administered VMware vSphere infrastructure (vCenter, ESXi) across hybrid environments
- Managed AWS workloads and Cisco Meraki cloud-managed networking
- Deployed and maintained Fortinet NGFW policies and site-to-site VPN infrastructure
- Supported Cisco and Aruba networking equipment across enterprise locations
- Implemented CrowdStrike Falcon endpoint protection across enterprise endpoints

### Freelance Consultant

*RYIO Consulting | Mar 2022 - Jun 2023*

- Provided IT consulting and infrastructure support for small and mid-sized businesses
- Managed Windows Server and Linux environments including Active Directory and Group Policy
- Supported AWS and Microsoft Azure infrastructure deployments
- Performed infrastructure troubleshooting, system upgrades, and maintenance

- Assisted clients with network configuration, system administration, and operational support

### Systems Administrator

*BlueAlly | Oct 2019 - Dec 2021*

- Managed Fortinet Security Fabric deployments, Cisco ASA, and Cisco Firepower firewalls
- Implemented Riverbed SteelHead WAN optimization appliances
- Deployed Talari SD-WAN solutions to improve connectivity across distributed sites
- Administered Linux servers, including package management, services, and automation scripts
- Maintained strict SLA requirements across managed enterprise environments

### Help Desk Specialist (Contract)

*Casey's General Stores | Jul 2019 - Oct 2019*

- Supported rollout of Casey's online ordering platform across retail locations
- Assisted with incident triage, ticket resolution, and remote troubleshooting
- Provided technical support for store systems and retail infrastructure
- Worked with deployment teams during rollout of new ordering systems

### Datacenter Technician (Contract)

*Microsoft | Nov 2017 - Dec 2018*

- Diagnosed and triaged server and hardware failures across datacenter clusters
- Performed Cisco Nexus switch configuration and network troubleshooting
- Assisted with rack deployments and server cluster provisioning
- Maintained compliance with strict Microsoft SLAs for outages and hardware failures

## SECURITY PROJECTS

### Bitprobe - Vulnerability Scanning Platform

*Personal Project | 2025 - Present | Tech: Python, Go, TypeScript, Node.js*

- Building a modular vulnerability scanner for web applications, network services, and TLS configurations
- Developing scanning modules to detect exposed services, TLS misconfigurations, and known CVE vulnerabilities
- Implementing service discovery and attack surface enumeration
- Building reporting output for vulnerability tracking and remediation
- Designed for security research, automated analysis, and penetration testing workflows

## EDUCATION

### Western Governors University

B.S. Cybersecurity and Information Assurance

*In Progress - Started January 2026*

### Des Moines Area Community College

A.A.S. Information Technology - Network Administration

*Graduated May 2016*

## CERTIFICATIONS

- ISC2 Certified in Cybersecurity (CC)
- LogicMonitor Certified Professional (Expired)
- Riverbed RCPE Associate (Expired)
- Fortinet NSE3 (Expired)